



Thesis Defense Guidelines

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Undergraduate Computer Science & Data Science

Presentation Deck & Slide Design

1. Core Content Structure

Your slide deck must include the following sections in sequence:

1. **Title Slide:** Project title, student names, and adviser's name.
2. **Background of the Study:** Context and motivation behind the research.
3. **Problem Statement:** The specific challenge or research gap being addressed.
4. **Objectives of the Study:** General and specific goals of the project.
5. **Conceptual Framework:** The theoretical or systemic overview of your solution.
6. **Methodology:** Algorithm design, system architecture, data pipeline, and evaluation metrics.
7. **Results and Discussion:** Performance analysis, experimental outputs, and data interpretations.
8. **Conclusions & Recommendations:** Summary of findings and potential future work.

2. Slide Design Best Practices

- **The 7x7 Rule:** Keep text concise. Limit slides to a maximum of 7 rows of text, with no more than 7 words per row.
- **Complementary Text:** Use short phrases on screen rather than cryptic fragments or long paragraphs. Your spoken words should expand upon the slide text, not duplicate it.
- **Visual Typography:** Emphasize core technical terms, variables, or metrics using **boldface** or *italics*.
- **Slide Numbering:** Include a slide number in the lower right-hand corner of every slide to help panelists reference specific content during Q&A.
- **Persistent Navigation:** Use clear slide headers to track which chapter or system phase is currently being discussed.
- **Data Visualization:** Rely heavily on high-quality diagrams, charts, system architectures, and flowcharts rather than text blocks to help the audience interpret your findings.

Oral Presentation Mechanics

1. Delivery, Timing, & Engagement

- **Time Limit:** Limit the absolute length of your total presentation to **30 minutes**. Rehearse extensively as a group to track how long each section takes so you stay strictly within bounds.
- **Introduction:** Begin by introducing your group members, your research adviser, and your exact thesis title.
- **Audience Contact:** Maintain direct eye contact with the evaluation panel. Avoid staring at the floor, the walls, or reading straight from the projection screen.
- **Pacing and Tone:** Monitor your tempo so you do not talk too fast or too slowly. Dynamically vary your voice modulation and pitch to emphasize critical breakthroughs or results.
- **Slide Mastery:** Show your slides properly. Do not flash slides on screen for only a second or two before moving forward; give the audience time to absorb the text.
- **Visual Explanations:** Actively walk the audience through your diagrams, system modules, or tables. Do not expect panelists to decipher complex visuals on their own.

- **Minimal Note Dependence:** Memorize your core delivery. Avoid reading word-for-word from scripts or flashcards during the presentation; notes should only be used as a rare, brief reference.

2. Technical Demonstrations (CS/DS Specifics)

- **Live Demos:** If your thesis involves software, a web app, or a trained machine learning model, allocate 3 to 5 minutes of your presentation time for a live demo.
- **Pre-Recorded Backup:** Always have a high-quality video recording of your system working successfully in case of sudden internet drops, environment errors, or hardware glitches during the live defense.
- **Code Preparedness:** Have your clean, well-commented source code open in an IDE in the background, ready to present if a panelist requests to see an algorithmic implementation.

3. Team Coordination & Logistics

- **Equal Speaking Distribution:** Divide the presentation topics evenly among all group members. Every student must present a core technical component of the study.
- **Smooth Transitions:** Practice passing the floor to your teammates professionally (e.g., *"Now, I will pass the floor to [Name] to discuss our experimental results."*).
- **Equipment Dry Run:** Arrive at the defense venue early to test your laptop's connection to the projector, check slide formatting on the screen, and ensure audio levels are clear.

Handling Panel Questions

- **Active Listening:** Listen completely to the panelist's query without interrupting. Ask for clarification or rephrase the question in your own words before attempting to answer.
- **Broadcast the Answer:** Speak clearly and project your voice to the entire room. The answer is meant for the benefit of the whole panel, not just the individual who asked it.
- **Maintain Professional Objectivity:** Accept academic critique gracefully. Remember that the panel is evaluating the current state of the project, not your personal intelligence or competence.
- **Honesty Over Evasion:** Do not attempt to deflect or hide project flaws. If your project has a known constraint or limitation, state it honestly. Panelists value methodological honesty over deceptive answers.
- **Address Every Question:** Treat every question from a panelist as highly relevant. If you believe a point is out of scope, explain your reasoning professionally and objectively rather than dismissing it.
- **Do Not Retaliate:** Remain calm, polite, and composed at all times. Defensiveness or sarcastic remarks will negatively affect your evaluation.